

[SINGLE CORRECT CHOICE TYPE]

Q.1 to Q.10 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

- Q.1 From the point (1, 1) tangents are drawn to the curve represented parametrically as $x = 2t - t^2$ and $y = t + t^2$. The distance between the points of contact, is
(A) $\frac{2\sqrt{43}}{9}$ (B) 2 (C) 3 (D) $\frac{2\sqrt{53}}{9}$
- Q.2 Let $f(x)$ be a function defined by $f(x) = \int_1^x (t^2 - 3t + 2) dt$, $1 \leq x \leq 3$. Then the range of $f(x)$ is
(A) $[0, 2]$ (B) $\left[\frac{-1}{4}, 4\right]$ (C) $\left[\frac{-1}{4}, 2\right]$ (D) none of these
- Q.3 If the eccentricity of the ellipse $3x^2 + 4y^2 = 12$ is changing at the rate 0.1/sec, then the time at which it will coincide the auxiliary circle, is
(A) 2 sec (B) 3 sec (C) 5 sec (D) 6 sec
- Q.4 Let f and g be two continuous function such that $g \circ f$ is increasing and g is decreasing. If $f(g(x^2 - 2kx + 3)) > f(g(k - 3)) \forall x \in \mathbb{R}$, then number of possible integral values of k is
(A) 3 (B) 4 (C) 5 (D) 6
- Q.5 For the function $f(x) = \int_0^{x^2 - t^2} e^{\frac{-t^2}{4}} (4 - t) dt$, which of the following statement is **incorrect**?
(A) a local maximum occurs at $x = 2$ (B) a local maximum occurs at $x = -2$
(C) a local maximum occurs at $x = 0$ (D) a local minimum occurs at $x = 0$
- Q.6 Given $f'(1) = 1$ and $f(2x) = f(x) \forall x > 0$. If $f'(x)$ is differentiable, then there exists a number $c \in (2, 4)$ such that $f''(c)$ equals
(A) $\frac{1}{4}$ (B) $\frac{-1}{2}$ (C) $-\frac{1}{4}$ (D) $-\frac{1}{8}$
- Q.7 If a variable tangent to the curve $x^2y = c^3$ makes intercepts a, b on x and y -axis, respectively, then the value of a^2b is
(A) $27c^3$ (B) $\frac{4}{27}c^3$ (C) $\frac{27}{4}c^3$ (D) $\frac{4}{9}c^3$

- Q.8 Given that $f(x)$ is continuously differentiable on $a \leq x \leq b$, where $f(a) < 0$ and $f(b) > 0$. Which of the following statement is **not true**?
- (A) $f(x)$ is bounded on $a \leq x \leq b$.
 (B) The equation $f(x) = 0$ has at least one solution in $a < x < b$.
 (C) The maximum and minimum values of $f(x)$ on $a \leq x \leq b$ occur at points where $f'(c) = 0$, $a < c < b$.
 (D) There is at least one point c with $a < c < b$, where $f'(c) > 0$.
- Q.9 The ordinates of the point on the curve $x^3 = y(x-a)^2$, ($a \neq 0$) where the ordinate is minimum, is
- (A) $\frac{27}{4}a$ (B) $8a$ (C) $3a$ (D) $2a$
- Q.10 The least integral value of k for $f(x) = \frac{e^{-x}}{\sqrt{k+8x^2}}$ is decreasing for all $x \in \mathbb{R}$ is
- (A) 1 (B) 2 (C) 3 (D) 4

[PARAGRAPH TYPE]

Q.11 to Q.15 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

Paragraph for question nos. 11 & 12

Let $f(x) = x^3 - 6 \sin \theta \cdot x^2 + (2 \sin^2 \theta + q)x - 1$, $q, \theta \in \mathbb{R}$ be a cubic polynomial in x . If exactly one tangent can be drawn to the graph of $y = f(x)$ which is parallel to the line $y = x$, then

- Q.11 the largest value of q is
- (A) 7 (B) 9 (C) 11 (D) 13

- Q.12 if q is smallest, then the value of $\left. \frac{d}{dx} (f^{-1}(x)) \right|_{x=9}$ is
- (A) $\frac{1}{2}$ (B) $\frac{1}{13}$ (C) $\frac{3}{17}$ (D) $\frac{2}{9}$

Paragraph for question nos. 13 to 15

Let $y = g(x)$ and $y = P(x)$ be two polynomial functions of degree two and four respectively such that

- (i) $P(-x) = P(x)$ and $P(x) \geq 0 \forall x \in \mathbb{R}$
 (ii) $P(x)$ has two local minimum at $x = x_1, x_2$ such that $|x_1 - x_2| = 2$.
 (iii) The line $y = 1$ touches the curve $y = P(x)$ at point of local maximum and encloses an area of

$$\frac{8\sqrt{2}}{15} \text{ sq. units with it}$$

- (iv) $\lim_{x \rightarrow 0} \frac{P(x) - g(x) - g(-x)}{x^2} = g'(-1)$.

- Q.13 Minimum value of $P(x)$ equals

- (A) 2 (B) 1 (C) $\frac{1}{2}$ (D) $\frac{1}{4}$

Q.14 If $g(x)$ has maximum value at $x = -1$ then product of its roots is

- (A) $\frac{-1}{2}$ (B) -1 (C) $\frac{1}{2}$ (D) 1

Q.15 $y = P(x)$ has inflexion points at

- (A) $x = \pm \frac{1}{\sqrt{3}}$ (B) $x = \pm \frac{1}{2}$ (C) $x = \pm 2$ (D) $x = \pm \frac{1}{\sqrt{2}}$

[MULTIPLE CORRECT CHOICE TYPE]

Q.16 to Q.19 has four choices (A), (B), (C), (D) out of which **ONE OR MORE** may be correct.

Q.16 Which of the following statements is(are) true for $f(x) = \begin{cases} \sqrt{x}, & x \geq 1 \\ x^3, & 0 \leq x < 1 \\ \frac{x^3}{3} - 4x, & x < 0 \end{cases}$?

- (A) $f(x)$ is monotonically increasing in $(-\infty, -2) \cup (0, \infty)$
(B) function attain its extremum value at x_1 and x_2 where $x_1 + x_2 = -2$
(C) function is not differentiable at two points
(D) function is discontinuous at one point

Q.17 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous and differentiable function such that $f(2) = 3$, $f(6) = 8$ and has local minimum at $x = 2$. If $g(x) = \int_2^x f(2x+t) dt$, then

- (A) $g'(0) = 5$
(B) $g'(0) = 10$
(C) $\lim_{x \rightarrow 0} \frac{g(f(x+2)-3) - g(0)}{x} = 0$
(D) $\lim_{x \rightarrow 1} \frac{f(6x) - f(6)}{x-1} = \lim_{x \rightarrow 0} \frac{6f(x+6) - 16f(x+2)}{x}$

Q.18 Consider a polynomial function $f(x) = x^5 + ax^4 + bx^3 + cx^2 + dx + e$. If the line $y = 2$ touches it at $x = 1$ and $x = 4$ where $x = 4$ is also a point of inflection of $y = f(x)$, then

- (A) at $x = \frac{11}{5}$, $f(x)$ has a local minimum. (B) at $x = \frac{11}{5}$, $f(x)$ has a local maximum.

- (C) the value of $f(2) + f(3)$ equals -8 . (D) the value of $\int_2^6 f(x) dx$ is $\frac{424}{5}$.

Q.19 If l and m are real number such that $x^3 - lx^2 + mx - 27 = 0$ has its roots real and positive, then

- (A) $l_{\min} = 3$ (B) $l_{\min} = 9$ (C) $m_{\min} = 8$ (D) $m_{\min} = 27$

[INTEGER TYPE]

Q.20 & Q.21 are "Integer Type" questions. (The answer to each of the questions are upto **4 digits**)

Q.20 If a differentiable function 'f' satisfies the relation

$$\int_0^t f^2(x) dx + \frac{1}{2} \int_0^{\pi/2} t^6 \sin^3 x dx = \int_0^t 2xf(x) dx \quad \forall t \in \mathbb{R}, \text{ then find the number of solution (s) of the equation}$$

$$f(x) = \sqrt[3]{x}.$$

Q.21 Let a cubic polynomial be $f(x) = \frac{x^3}{3} - x^2 + ax + 2$. If all the values of a for which $f(x)$ has a positive point of maximum also satisfy the inequality $3x^2 - (b+1)x + b(b-2) < 0$ and value of b lies in $[p, q]$, then find the value of $(p+q)$.

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ANSWER KEY

Q.1	D	Q.2	C	Q.3	C	Q.4	B	Q.5	C
Q.6	D	Q.7	C	Q.8	C	Q.9	A	Q.10	B
Q.11	C	Q.12	B	Q.13	C	Q.14	B	Q.15	A
Q.16	ABC	Q.17	BCD	Q.18	ACD	Q.19	BD	Q.20	2
Q.21	3								

